

MDM Mini Project Report

StudXchange – Student Exchange and Marketplace System

Team Members

Sr. No.	Team Member
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Problem Statement

Students often have unused books, notes, electronics, and other items but lack a proper platform to exchange them within campus. This leads to wastage of resources and limited communication among students.

Description

StudXchange is a web-based platform that allows students to buy, sell, and rent items within their campus community. It includes real-time chat functionality for direct communication between users, a Committee Hub for college clubs and events, and a Rent & Borrow section for short-term item usage.

Technology Used

Layer	Technologies
Client Side	HTML, CSS, JavaScript
Server Side	Node.js, Express.js, Socket.io
Database	MySQL

Layout Design / Snapshots

Home Page

The Home Page is the main landing page of StudXchange. It showcases the platform's core value proposition and gives quick access to Marketplace, Rent & Borrow, and Committee Hub. It explains how the platform works in three steps and highlights VCET-only access, always-free usage, and full device responsiveness.

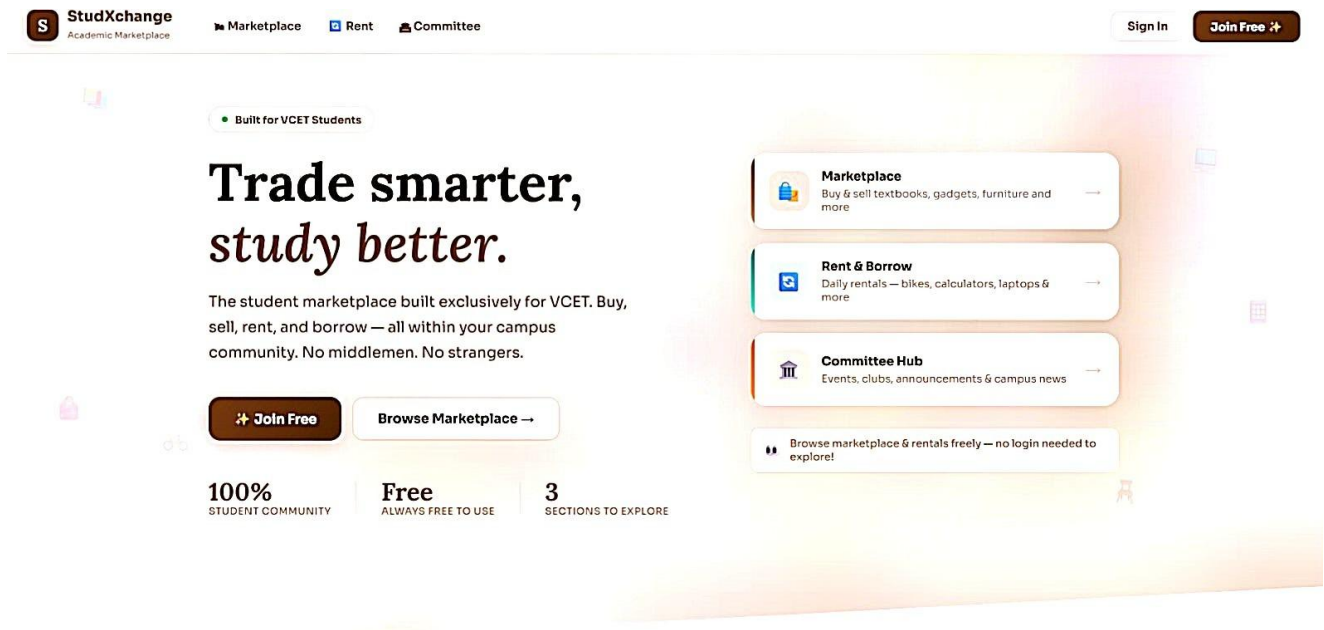


Fig 1.1 – Hero Section (Trade smarter, study better)

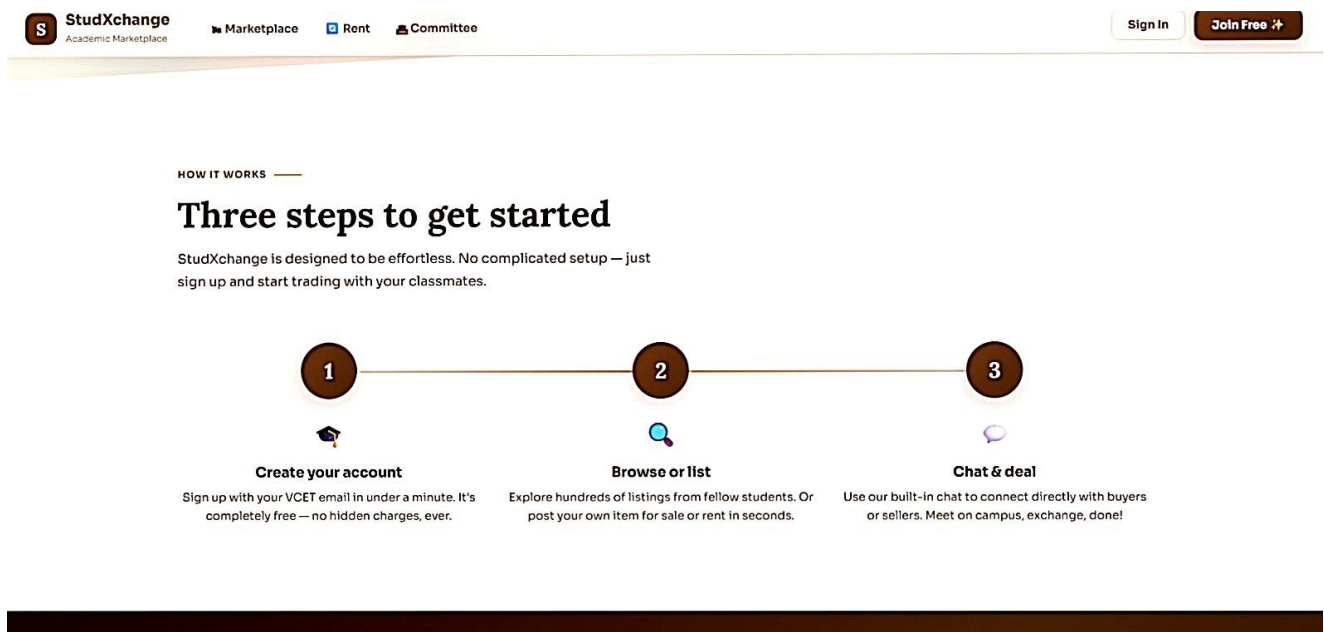


Fig 1.2 – How It Works (Three steps to get started)

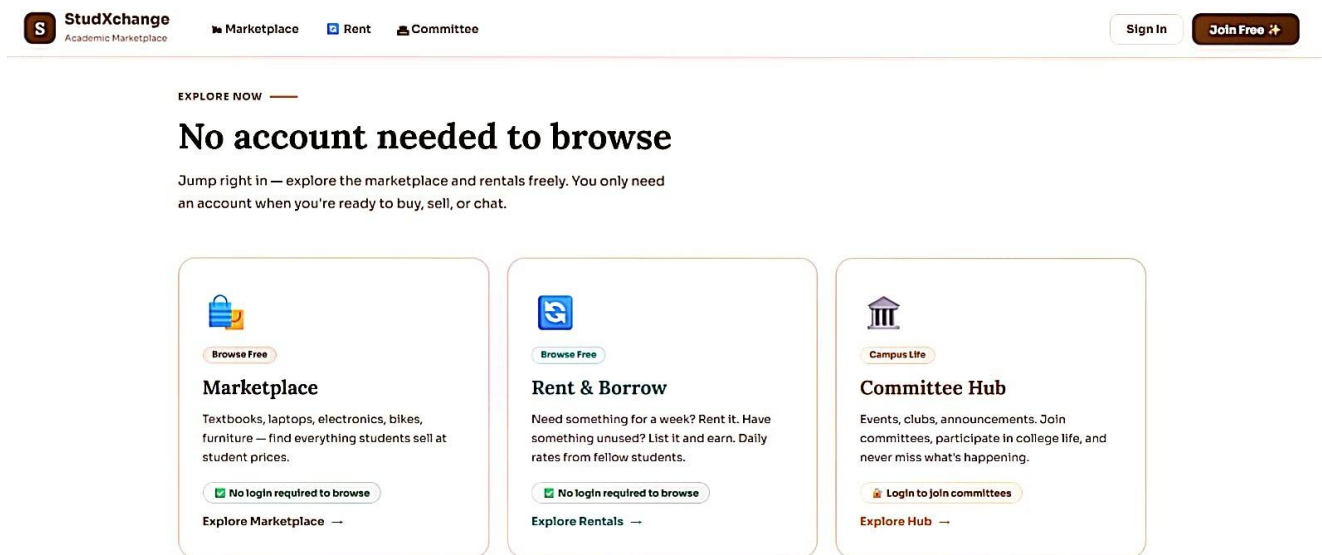


Fig 1.3 – Explore Sections (No account needed to browse)

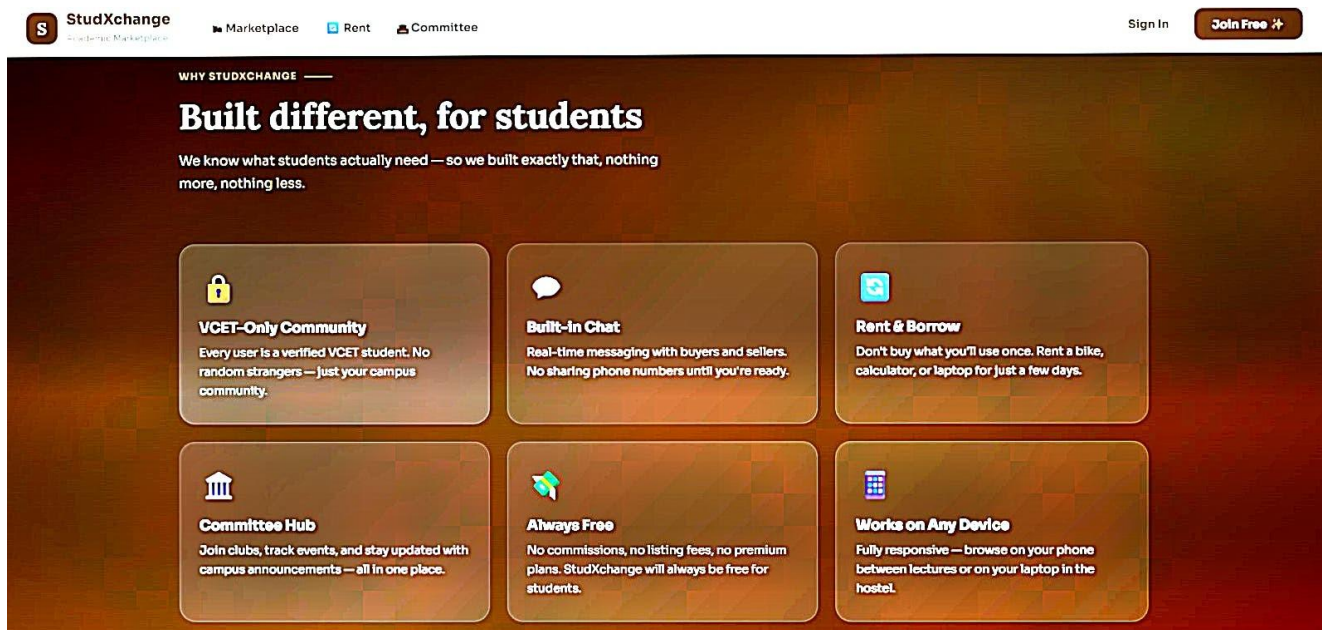


Fig 1.4 – Why StudXchange (Built different, for students)

Login / Register Page

Allows VCET students to sign in using their college email. A split-screen layout reinforces branding on the left and presents the sign-in form on the right. New users can create an account directly from this page.

Marketplace Page

Central hub for buying and selling. Students browse listings by category (Laptops, Books, Electronics, Notes, Bikes, Furniture), sort by newest first, and post new items directly from this page.

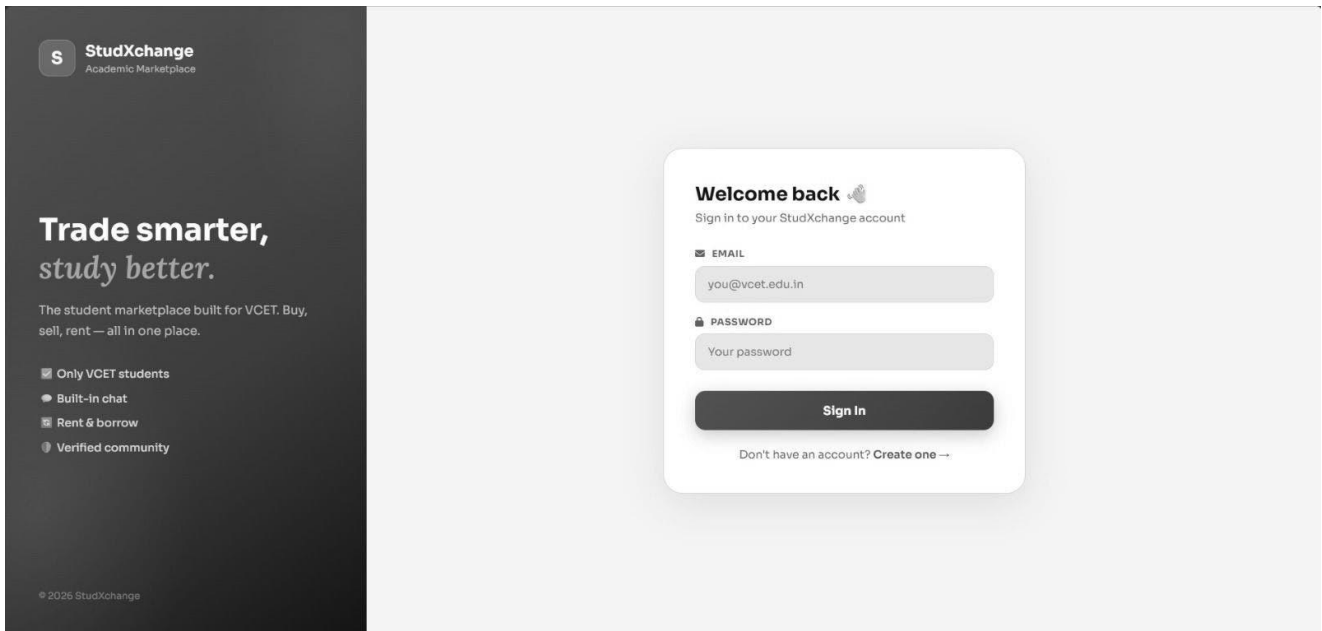


Fig 2 – Login / Register Page

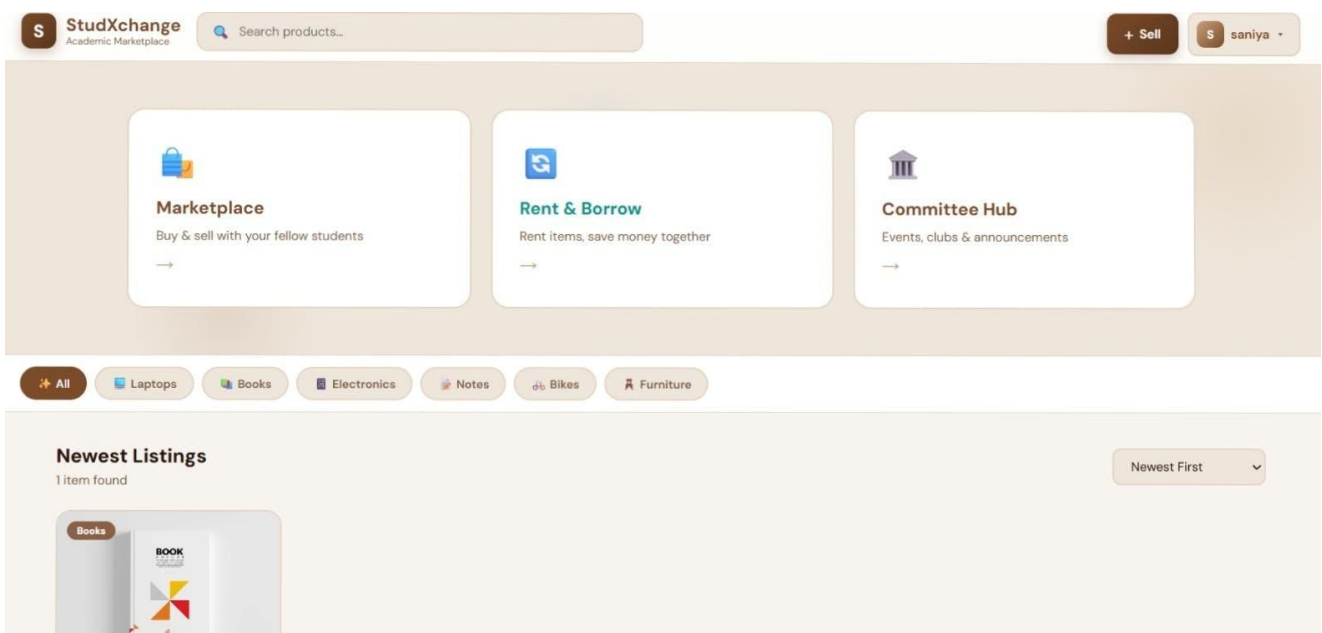


Fig 3 – Marketplace Page

Product Details Page

Displays full item information including title, price, category, seller, condition, and description. Buyers can initiate contact with the seller via built-in chat.

Sell / Post Item Page

Lets students list items for sale by filling in title, price, category, condition, description, and an optional photo.

Designed for quick listing with minimal friction.

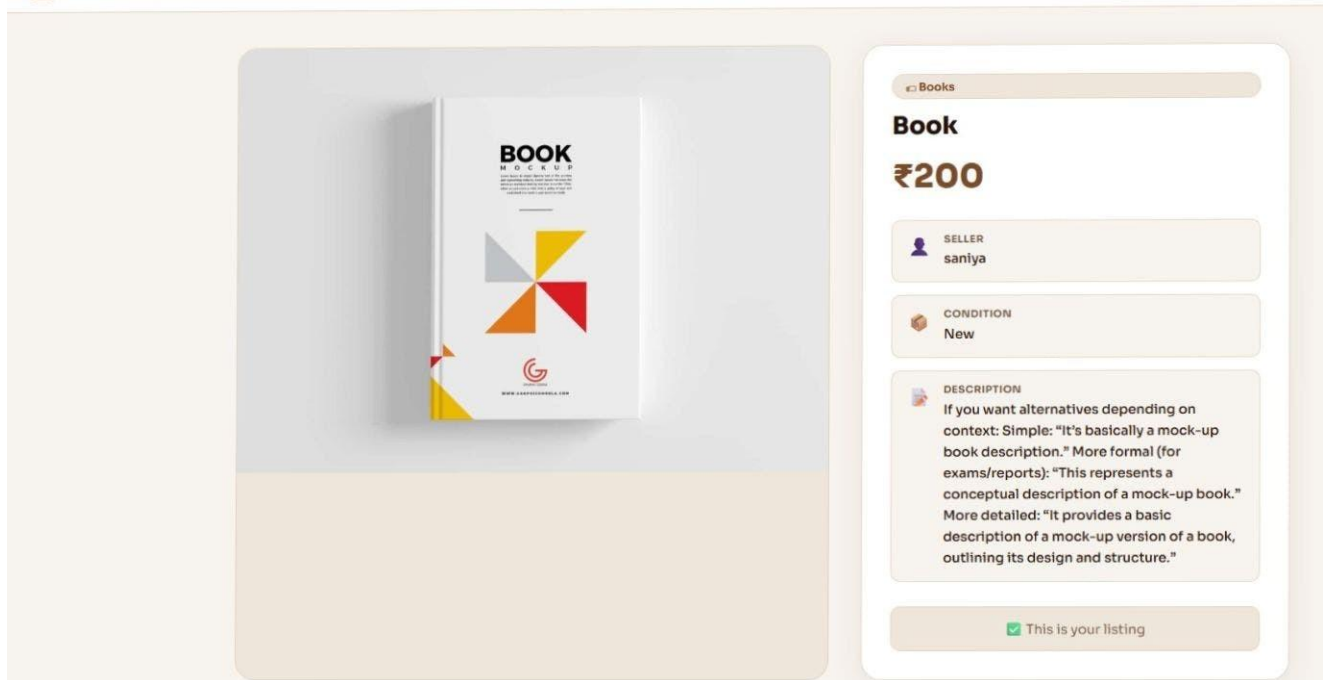


Fig 4 – Product Details Page (Buy)

**Post Your Item**
Fill in the details to list your item for sale

UPLOAD PHOTO


Click to upload image
JPG, PNG up to 5MB

TITLE *

e.g. Engineering Textbook

PRICE (₹) *

e.g. 350

CATEGORY

Books

CONDITION

New

DESCRIPTION

Describe your item — age, usage, any defects...

CONTACT NUMBER

+91 98765 43210



Fig 5 – Sell / Post Item Page

Rent & Borrow Page

Students rent out items they own or borrow what they need at student-friendly rates. Categories include Books, Laptops, Electronics, Bikes, and Furniture.

Chat Interface

Real-time messaging using Socket.io. Each conversation is tied to a specific listing. Users connect directly without sharing personal phone numbers.




No rentals yet
 Be the first to list something for rent!

Fig 6 – Rent & Borrow Page

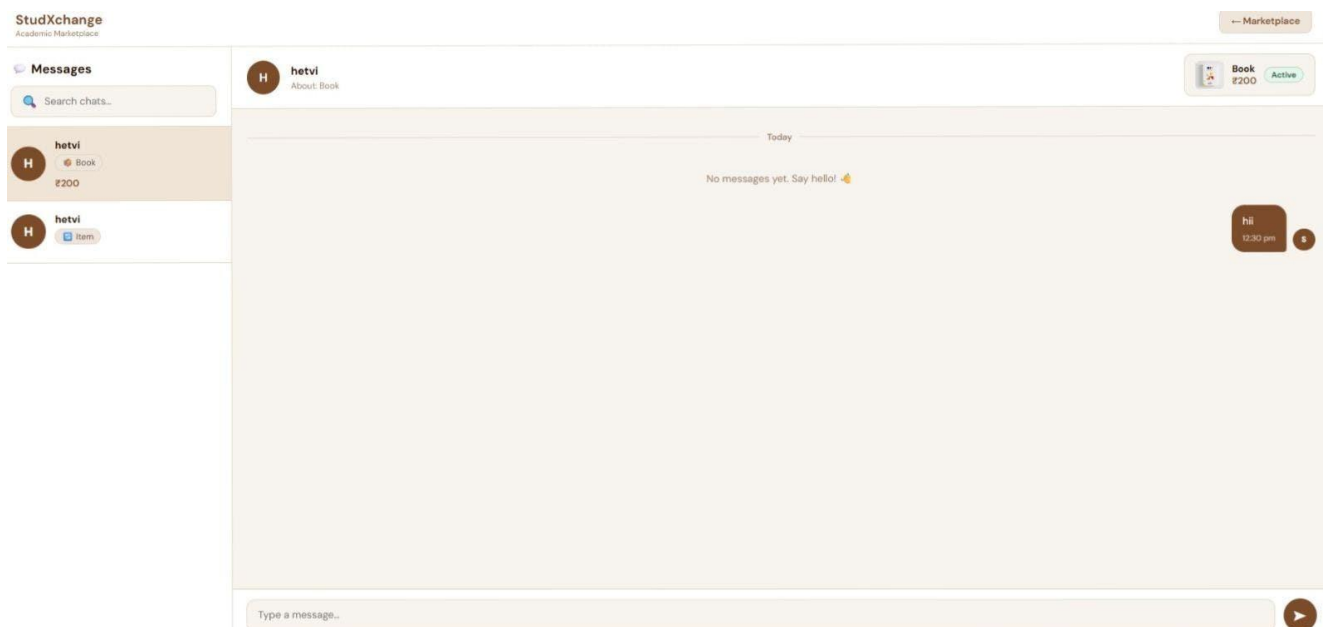


Fig 7 – Chat Interface (Real-Time Messaging)

My Profile Page

Displays a student's personal info, active listings, incoming requests, and wishlist. The Verified badge confirms VCET student identity.

Committee Hub

Dedicated section for campus life — discover and join committees like Coding Club, Cultural Committee, NSS, Sports Club, Photography Club, and Entrepreneurship Cell. Also tracks events, announcements, and activity feeds.



saniya

Student



Verified

Share

About Me

Edit



NAME

saniya



EMAIL

saniya.2405606209@vcet.edu.in



PHONE

—



COLLEGE

—



COURSE & YEAR

—



LOCATION

—



BIO

—

My Listings

Requests

Wishlist

ITEMS YOU LISTED FOR SALE OR RENT



Sale

Book

₹200

Books

Edit

Delete

Fig 8 – My Profile Page

NAVIGATION

Activity Feed

Committees

Events

My Groups

Notifications

MY COMMITTEES

Coding Club

All Committees

Explore and join your college committees

Search committees...

All Categories



Coding Club

TECHNICAL

Hackathons, coding contests, and workshops.

2 members

View



Cultural Committee

CULTURAL

Festivals, dance, music, and drama events.

0 members

View



NSS

SOCIAL SERVICE

Community development and volunteer work.

0 members

View



Sports Club

SPORTS

Inter-college tournaments and fitness challenges.

0 members

View



Photography Club

ARTS

Photo walks, editing workshops, exhibitions.

0 members

View



Entrepreneurship Cell

TECHNICAL

Mentoring, funding guidance, pitch competitions.

0 members

View

Fig 9 – Committee Hub (All Committees)

Official College Hub

Committee Hub

Join clubs, attend events, collaborate — your college life, organized.

6

COMMITTEES

0

EVENTS

NAVIGATION

Activity Feed

Committees

Events

My Groups

Notifications

MY COMMITTEES

Coding Club

All Committees

Explore and join your college committees

Search committees...

All Categories



Coding Club

TECHNICAL

Hackathons, coding contests, and workshops.



Cultural Committee

CULTURAL

Festivals, dance, music, and drama events.



NSS

SOCIAL SERVICE

Community development and volunteer work.

Fig 10 – Committee Hub Landing Page with Stats & Navigation

Data Base Design

The database is designed using MySQL and follows a relational model. The ER Diagram below illustrates the key entities — User, Item, Message, Committee, Committee Member, Wishlist, and Rent Request — along with their attributes and relationships. Primary Keys (PK) uniquely identify each record, while Foreign Keys (FK) maintain referential integrity across tables.

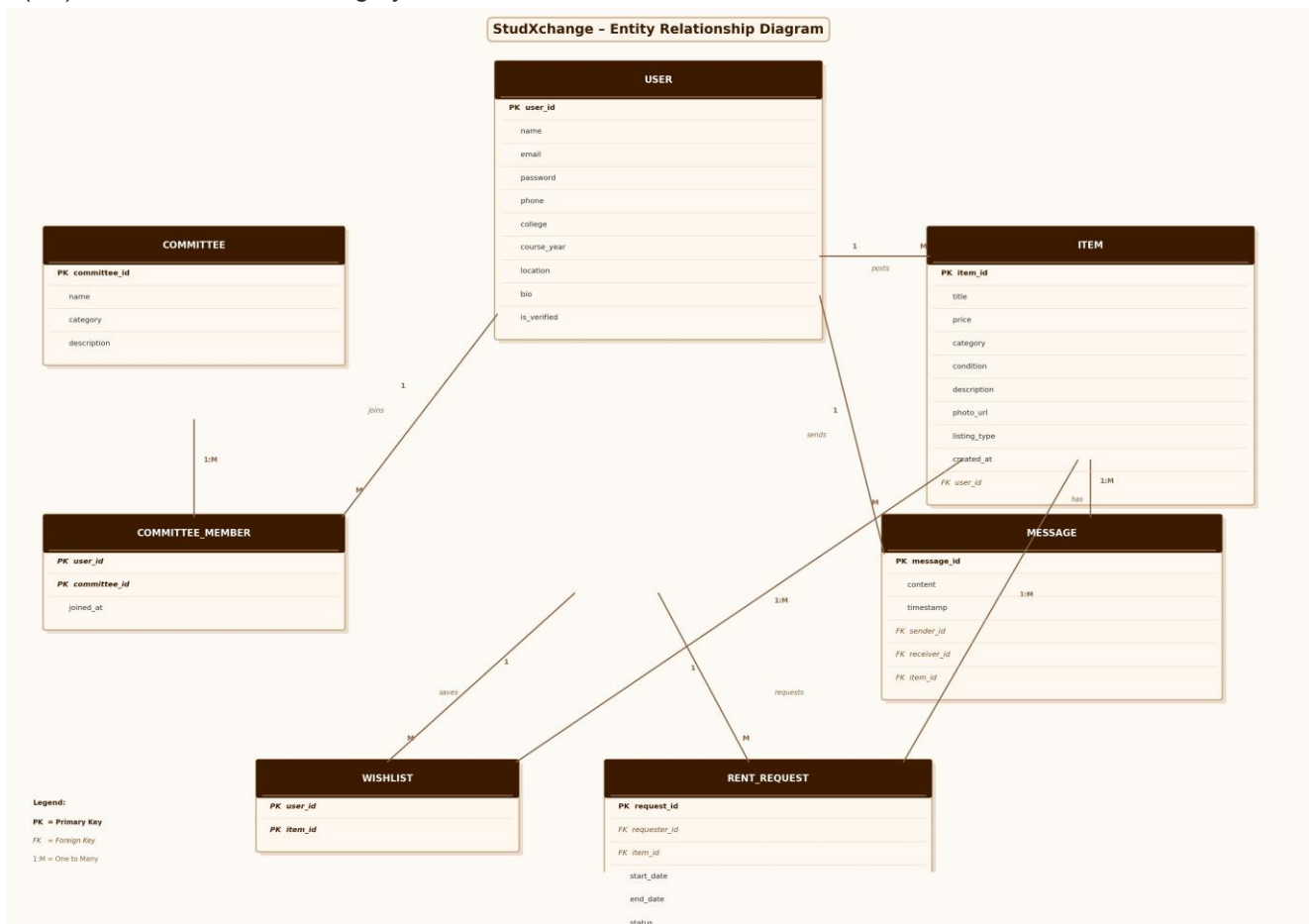


Fig 11 – Entity Relationship Diagram (StudXchange Database)

Conclusion

StudXchange provides an efficient, student-first platform for exchanging resources within a campus community. By combining a buy/sell Marketplace, a Rent & Borrow system, real-time chat, user profiles, and a Committee Hub into a single web application, the project demonstrates full-stack development skills using modern technologies. The platform reduces resource wastage, eliminates the need for third-party marketplaces, and fosters stronger communication and collaboration within the student community.